

Foreword

Where Will Math Lead You? Over 50 Years of David Harbater's Mathematical Journey and Insight

Professor David Harbater, Class of 1970, is a mathematics professor at the University of Pennsylvania and is most known for his work in Galois theory, algebraic geometry, and arithmetic geometry. While attending Stuyvesant, he was an editor for Math Survey and a co-captain of the Math Team. After graduating from Stuyvesant, he attended Harvard College and graduated summa cum laude in 1974. Harbater then earned a master's degree from Brandeis University followed by a Ph.D. from MIT in 1978, where he wrote a dissertation (Deformation Theory and the Fundamental Group in Algebraic Geometry) under the direction of Michael Artin. In 1995, Harbater was awarded the Cole Prize along with Michel Raynaud for their solution of the long-outstanding Abhyankar conjecture.

This foreword is based on an interview conducted by Josephine Lee. The content has been edited for clarity.

“The beauty of human creativity” at Stuyvesant

I was involved in various math activities at Stuyvesant, with one of them being an editor for *Math Survey*. It was a very different and tedious process since we had to use a typewriter to type the mathematical symbols. With the rise of new technology, people can now concentrate more on the mathematical aspect and less on the technical aspects. We used to go around to classrooms at 10 AM to make announcements saying, “We’re selling the *Math Survey*. So if you’d like to buy a copy, I have a whole bunch of copies here.” I enjoyed the camaraderie of people staying around after school, typing up, and pasting things down.

We had three co-captains of Math Team: me, Arthur Rothstein, and Julius Collins. We were usually working on one area of math and relating it to other areas of mathematics to have a broad mathematical perspective. We saw how math relates to science and to everything else. It’s the beauty of human creativity.

The Hampshire College Summer Studies in Mathematics program, which I learned about through the Math Department, gave me a broader sense of mathematics and a taste of all the math up through college graduation. That was the first time I had the idea of what it meant to be a mathematician after conversing and engaging with professors. It led the way to both college and graduate school. In later summers, I attended the Ross Mathematics Program, first as a participant and again as a counselor.

The Stuyvesant math teacher that influenced me the most was Mark Schwartz. He pushed me to attend NYU for math classes and encouraged me to join the math team and apply to many summer programs. He also advised me to learn how to read math in French and Russian, since mathematics is international.

During my senior year, I studied French for one year, and it was useful when I was traveling to different conferences. It wasn't necessary just to read math, although I do have to read in French all the time because if I'm looking at something written 50 years ago, it might very well be in French.

Communicating and Collaborating

Being a mathematician has a collaborative nature, and it's becoming more collaborative as time goes on because of advancements in technology. When I was getting my Ph.D., most of the articles were published or written by one person, with only some very important articles written by two. And rarely I would see one written by three. These days, most papers have more than one author.

For instance, I am working on finishing one paper, and there are six of us with various perspectives writing from different places. We have grown up on different continents, making it a very diverse group. We communicate with each other well through e-mail and Zoom, which to some extent is solitary. However, after you come up with something and then you tell the others, it bounces back and forth and it leads to much more productivity than you could accomplish otherwise. Individually, a single person could not have achieved it, but together you can.

Although some say that mathematics is mostly done by young people before 30, I've found that's not really true. I love the fact that I was about 40 years old when I won the Cole Prize, and that I've also done significant work since then, including getting published into a very top journal.

Another great aspect of collaborations is that I can work closely with people with both older and younger perspectives: the perspective that includes concepts from 20 years ago and also the other perspective of newcomers to the field thinking about it for the first time who are forming their own ideas and coming to it with fresh eyes.

You usually think mathematics is learned from books, but it's really an oral tradition with an interpersonal aspect. People learn how others think about math, especially their Ph.D. thesis advisors, and that advisor learned that thinking from their advisor, et cetera. You're sort of carrying along this long-term tradition of people thinking in a certain way, so you feel like a part of history. Of course, mathematical thinking evolves over time. But understanding the history often reveals the motivation for that line of thinking.

“Pushing civilization forward” with this “human endeavor”

Mathematicians try to understand phenomena by solving problems and answering questions. Every question has consequences; it's not just a random inquiry. We often think of mathematics as this perfect crystal that exists separate from human existence or even the universe. But mathematics is actually a product of human beings thinking outside our own experience. Mathematical thinking originated from human curiosity, which is an important distinction. Abstract mathematics is filled with all these definitions because there was something they were trying to understand; they were trying to answer a question. So mathematicians should always ask: Why is this concept really important?

When I was a grad student, my Ph.D. thesis advisor taught this to me about math research: You're not just trying to find something that's true or even something that no one ever said before. That's not enough. Here's an example: You write down a 50-digit number. Underneath it, you write another 50-digit number. You draw a horizontal line, and you add them. And the new statement is this number plus this number equals another number. That is a mathematical statement. It's true. If you didn't make any mistakes, no one ever said it before.

But it is not interesting, so it does not count as mathematical research. But what's the mathematical definition of interesting? It needs to deal with people. So that's the weird aspect of mathematics. On the one hand, it has this pure aspect that exists on its own. On the other hand, it's being done by people and people have to find it interesting. It won't make its place on *Access Hollywood*, but it's interesting in some sense of the word.

I think of research as contributing to the long-term process of pushing civilization forward. And to me, that's a high purpose of human beings, to move closer to something greater than themselves. I don't know what that thing is, but good things happen when people create and do things that others appreciate.

Are You Willing to be Wrong? Are You Willing to Fail?

Another professor in my department says that doing mathematics is often just like banging your head against the wall, and you keep banging in hopes that the wall cracks first. When trying to create new mathematics, most ideas that you have don't work. But the ideas that do work will contribute and move you forward a little bit. Then you try to do the next step. And again, you're stuck, and you keep going until you get the next thing that works. So it's a long-term process. You definitely need patience, and you need to be willing to make attempts that don't work. If you're going to do math research, you have to like it. If you don't like it, it's definitely not going to work.

Mathematicians must remember that truth is not just we want it to be. Outside of math, if the truth is inconvenient, they can ignore it. But in mathematics, if you don't have the mindset of "I'm going to find everything that's wrong with my research," someone else will at a later point, maybe even after it's published. You must be willing to work on a problem or conjecture for a long time, and still be willing to have other people tell you that you're wrong and to find that you're wrong yourself.

In coursework, students easily get a misleading idea about mathematics. They hear about all the successes, and they think that everything is known, but in reality, most things are not known. It's just that we don't spend a lot of time in classes telling people, "Oh, we don't know this." So students should be prepared to work on problems which are very difficult. When I was thinking so hard about a certain problem, I felt like my brain was going to explode, but I eventually managed to solve it, and then I got a big prize for it, which was very nice because I felt like I was contributing. On the other hand, I'm still thinking about an even bigger problem that I started working on even earlier. Maybe I'll never solve it, maybe someone else will solve it using my strategy, or maybe they'll combine it with some other strategy and solve it. If my work helps one way or another, that's great.

Learning How to Learn

Too often in math, we don't convey what things are fundamentally about. Textbooks just write down what we know in a very formal way. I tell students they do have to learn math from textbooks, but that's not all. They also have to understand the ideas, the part that's supposed to stay in their heads, and realize that mathematics is not just what's written down. And so I enjoy the challenge of teaching students how to think about what mathematics is really all about and how to use their intuition.

Intuition is crucial. Without it, you won't know how to work with the concepts or what questions to ask when doing research. Although intuition is necessary, it's not sufficient because you also have to know the official stuff that gets written in books.

But if no one is telling you what the sort of “secret part” is, the part that’s not written down, then you have to try to come up with it for yourself. What you come up with might be what other people are also thinking of, or it might be something new which might actually be really good. In fact, some of my research emerged from this process: taking things that were written down and asking, “Well, what’s really going on?”

The Past, Present, and Future of Math

In the past, mathematics and physics were not distinct fields of study; they were parts of natural philosophy. After they divided, mathematics separated into various branches such as algebra, geometry, etc. But this sub-categorization proved to be counterproductive because mathematicians were building mountains in the center of each of these areas and leaving valleys in between with questions that no one was looking at. Making progress in the middle of a specific area is like taking a rock and carrying it to the very top of the mountain. But with less work, you could move a whole bunch of rocks into a valley and build a really interesting structure, collaborating and merging fields.

What counts as applied mathematics, and where the boundaries lie between different areas, may become even fuzzier as mathematicians and scientists work together more. Very different disciplines might interact with each other in ways that will help each other, creating a richer structure than we have now.

The questions that form the field of mathematics will also change based on what people find interesting, such as abstraction versus not abstraction. For a while, the trend has been going more and more abstract, but in reaction, some mathematicians have moved towards working on more concrete applications. In the future, more contrasting areas of study will be able to coexist.

Mathematics keep changing because it is influenced by what’s happening in the world. As mathematicians encounter certain questions they didn’t consider before, then they will work on them. For example, I can imagine some area of mathematics coming out of pandemics. Then, that branch of mathematics will take on a life of its own, independent of where it emerged from.

Studying math can lead you to work in all sorts of disciplines - computer science, physics, economics - because it trains your brain and forces you to think very precisely. It lets you focus on how to analyze problems and find structure. Math is more about the structure and method of thinking than it is about numbers.

Many wonder about the extent to which computers rather than humans will be doing math. Computers already work with examples in ways that humans couldn’t have before, and in the future, computers will learn more and more by examples rather than definitions. Mathematicians and computer scientists have also been able to write programs to tell the computer how to break proofs down into cases and then handle each case specifically. And then there’s even the question of whether computers will be taught, not just how to prove things, but even how to pose interesting questions, which is a complicated process.

Technology will continue to influence mathematics, not just because machines can perform computations, but because technology enables communication and collaboration. Mathematics increasingly welcomes all different kinds of people, even though it didn’t before. Smart people who are interested in math can do good in the world, which makes the other aspects comparatively irrelevant.

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